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Predicting Student Adaptability Level Using ML Models and XAI Approaches

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***Abstract*—As online learning becomes more prevalent, it is crucial to identify students who may struggle with adaptation. Existing machine learning approaches face three limitations: class imbalance in the dataset, manual hyperparameter selection leading to inferior performance, and black-box predictions that are difficult for educational stakeholders to interpret. This paper presents an adaptive prediction framework that combines automated optimization and explainable AI. The framework uses KMeansSMOTE to address class imbalance, Bayesian hyperparameter optimization to automatically tune model parameters, and dual-phase SHAP analysis to interpret model predictions. The proposed framework improves predictive performance while providing interpretable explanations for model decisions. The optimized Random Forest algorithm achieved 92.9% accuracy on a Bangladesh student adaptability dataset.**

***Index Terms*—Explainable AI, student adaptability, Random Forest, KMeansSMOTE, Bayesian optimization, SHAP**

I. Introduction

The shift from traditional learning to online learning, accelerated by pandemics, geopolitical crises, and natural disasters, has revolutionized educational delivery, but exposes variability in student adaptability during online learning [1]. Developing countries face limited infrastructure, limited device access, and financial constraints, posing serious obstacles to online learning.

Traditional statistical methods and modern machine learning (ML) approaches are the two main types of techniques that can be used to understand student adaptation in online learning environments. Traditional techniques rely on statistical analysis and survey-based assessments such as cognitive models with human-defined assumptions and often assume linear relationships [2].

Modern ML techniques such as Random Forest (RF), XGBoost capture complex relationships in student adaptability [3].

Identifying at-risk students is critical but current approaches rely on manual hyperparameter tuning, resulting in suboptimal performance and inconsistent reproducibility and black-box machine learning predictions (limited interpretability).

Unlike prior studies that apply only machine learning models directly to raw categorical features, our pipeline integrates (1) interaction-based feature engineering, (2) KMeansSMOTE for safe class balancing, (3) Bayesian optimization for automated tuning, and (4) dual-phase SHAP analysis to examine feature importance shifts before and after balancing. This combination enables the model to capture contextual relationships (socio-economic status, technological infrastructure, learning environment) and provide actionable insights for decision-making.

Specifically, the pre-processing phase applies KMeansSMOTE to address class imbalance, while the modeling phase uses Bayesian optimization to automatically tune model hyperparameters. Finally, dual-phase SHAP analysis before and after oversampling examines feature importance shifts. The proposed framework offers significant advantages in terms of model interpretability, automated optimization, and balanced learning.

Our paper is organized as follows: Section II presents related work, followed by Section III which presents proposed solution. Section IV contains results, and Section V presents conclusion and future work.

II. Related Work

Suzan et al. [4] established a benchmark for predicting student adaptability in online learning using a dataset of 1,205 samples from Bangladesh, achieving 89.63% accuracy with RF while highlighting the socio-demographic factors without addressing class imbalance.

Nnadi et al. [5] used the above models and integrate explainable AI (XAI) techniques such as (SHAP) and (LIME), achieving 91% accuracy of RF model. The result from SHAP analysis identified Class Duration as the most influential feature. Although interpretability was significantly improved, the study did not address class imbalance.

Eid and Raju [6] presented a comparative study combining machine learning models with advanced statistical analysis to explore how different adaptability features are connected. They used a method called Copula-based analysis.

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Colins et al. [7] tested multiple classifiers. They found that attributes like age, financial condition, institution type, location, and load-shedding conditions play an important role in determining student adaptability, without class imbalance.

Kar et al. [8] and Iparraguirre-Villanueva et al. [9] they used machine learning models with explainable AI (XAI) approaches to predict student adaptability, to make sure that applied models they used are transparent and easy to understand from educators.

A recent study [10] used RF as machine learning model with feature engineering as pre-processing for adolescent adaptability, without optimization or class imbalance handling. This limitation motivates our proposed approach [11].

Wongvorachan et al. [12] demonstrated that SMOTE enhances oversampling and undersampling techniques by improving minority class representation and reducing majority bias.

Literature reveals three key shortcomings: First, Class imbalance is ignored, allowing models to achieve high accuracy but fail to identify at-risk minority students. Second, Hyperparameter tuning is done using a manual grid or random search instead of using efficient Bayesian optimization. Third, Black-box models lack interpretability. Explainable AI (XAI) approaches such as SHAP analysis are needed to provide transparency. So this paper addresses all these problems. For class imbalance, we apply KMeansSMOTE to improve minority class. For manual tuning, we use Bayesian optimization which increases accuracy from 89.6% to 92.9%. For black-box predictions, we use SHAP analysis to explain model decisions while providing actionable insights for educators.

III. Proposed Solution

A. System Architecture

We propose an adaptive prediction pipeline to predict and assess student adaptability level in online education systems as shown in Fig. 1, Phase I (Pre-Processing) begins with data preparation, exploratory data analysis, and feature engineering followed by KMeansSMOTE oversampling technique. The pre-processed data is then passed to modeling stage. The Random Forest model (RF) is trained and optimized through hyperparameter tuning with cross-validation providing predictions for three adaptability levels. Phase II (Post-Processing) applies explainable AI techniques to interpret predictions using SHAP (Shapley Additive Explanations).

B. Phase I:Pre-Processing Phase

1) Exploratory Data Analysis (EDA): EDA was applied to visualize student adaptability dataset which enables the model to learn from individual predictors and interaction features that provide a full understanding of the conditions that shape student adaptability. The dataset is biased towards majority class and therefore unbalanced: Moderate

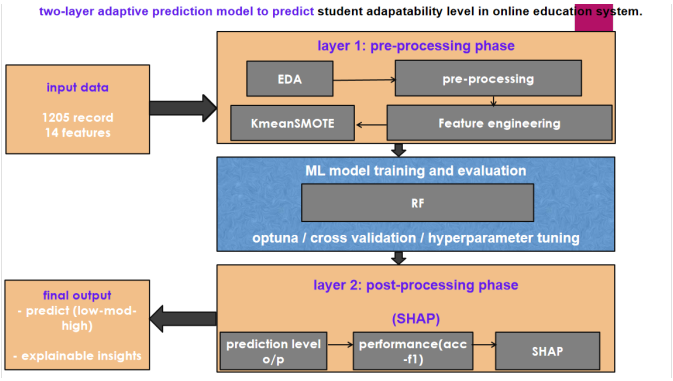


Fig. 1. Proposed adaptive prediction pipeline

(51.9%), Low (39.8%), High (8.3%) adaptability, Looking at the 13 categorical features, we organized dataset into four categories: - Demographics: such as gender, age and institution type.

- Educational context: describe students' learning context including education level, class duration, Learning management system (LMS) and IT students.

- Socioeconomic: students with financial conditions, location. these features help to identify students who face economic restrictions to learning.

- Technological Infrastructure: important features for understanding the technological environment that make students engage with online learning such as device, network type, internet connection type, and electrical stability through load-Shedding.

2) Data Pre-processing: All categorical features were transformed using one-hot encoding, which converts each nominal category into binary vectors. One-hot encoding was selected due to its compatibility with tree-based models and its ability to preserve categorical independence without introducing ordinal bias or target leakage. For missing value handling, we implemented mode-based imputation as a safeguard for categorical features and median-based imputation for numerical features. Stratified sampling (80/20) was used to split the data in order to ensure fair evaluation before making oversampling.

3) Interaction Feature Engineering: Raw categorical features miss contextual relationships. It creates interactions between features to capture complex relationships with variables, interaction terms were constructed following feature engineering best practices [13] to model contextual relationships beyond individual feature effects. So, we created three new interaction features via concatenation:

- **Tech-Access Index:** Internet Type + Device Type, 12 categories
- **Socio-Economic Status Index:** Financial Condition + Location, 6 categories
- **Engagement Profile:** Education Level + Self-LMS Usage, 6 categories

All categorical features, including both original and

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engineered features, were transformed using one-hot encoding to preserve interpretability and avoid ordinal bias.

4) KMeansSMOTE technique: We applied this method for the following reasons: KMeansSMOTE provides safe synthetic sampling because clustering ensures synthetic minorities are generated within homogeneous regions, avoiding noisy borderline areas that distort feature relationships this is evident in our results, where meaningful interaction features *Device_Internet_Mobile_Wifi* emerged to top four important features.

KMeansSMOTE handles between classes and within class imbalance, ensuring synthetic samples are generated where minority instances are sparsely distributed rather than uniformly distributed across feature space.

All of the above match with our objectives: improving minority class detection for high adaptability by achieving 95% while preserving the interpretability of core predictors like financial condition and class duration.

C. Learning Model Selection

We selected Random Forest (RF) as the base predictive model for three reasons. First, accuracy: Random Forest achieved 89.63%, compared to XGBoost (89.1%) and CatBoost (88.6%) [4], [9] Second, Robustness against Overfitting: By generating multiple trees, Random Forest is less prone to overfitting, as it averages the results and reduces the variability of predictions. The main advantage of RF is that trees protect each other from individual errors and when the model is trained on KMeansSMOTE generated synthetic samples where individual tree instability could propagate noise. [14].

Third, interpretability: Random Forest uses tree structure that enables SHAP-based explanations for actionable features such as class duration, financial condition) [10].

Models like XGBoost, which provide comparable accuracy but operate as black-boxes that are hard to interpret. RF strikes the optimal balance between predictive performance and explainability with explaining its reasoning through meaningful tree paths, making it the practical choice for our study. finally, in educational context, a model with slightly lower accuracy with transparent, interpretable reasoning is better than model achieving high accuracy black-box model. Interpretability is essential for stackholders to understand why a student is flagged as at-risk to design effective interventions. so unexplained high accuracy offers limited actionable visions for policy-making.

D. Phase II: Post-Processing Phase

1) Post-Hoc Explanation: The second phase of our proposed solution framework employs SHAP (Shapley Additive Explanations) to explain what the Random Forest (RF) predictions actually mean, SHAP values quantify feature contributions, enabling:

- **Global explanations:** it provides global Feature importance rankings across the dataset. This will produce full explanations for all data.
- **Local explanations:** provides a personalized explanation for individual student that highlights specific challenges and suggest ways of supporting.

This transforms black-box predictions into transparent, actionable intelligence for educators.

IV. Implementation And Experimental Results

A. Experimental Setup

We updated student data from the different levels (university, schools, and colleges) 1,205 samples in Bangladesh as a case study [4], available on Kaggle (free registration required) [15] The target variable is adaptability level, Which is a multi-class variable with three categories (High, Moderate, Low). We used 80% of the data as a training set and the remaining 20% as a test set. Feature descriptions are provided in Table I. All experiments were conducted using Python 3.9 with scikit-learn, imbalanced-learn, and SHAP libraries.

TABLE I
Dataset Features

Feature	Type	Categories
Gender	Cat	Girl, Boy
Age	Num	1–5, 6–10, 11–15, 16–20, 21–25, 26–30, 30+
Education Level	Cat	School, College, University
Institution Type	Cat	Non-Govt, Govt
IT Student	Bin	No, Yes
Location	Cat	Rural, Urban
Load-shedding	Cat	Low (83.3%), High (16.7%)
Financial Condition	Cat	Poor (20.1%), Mid (72.9%), Rich (7.1%)
Internet Type	Cat	2G, 3G, 4G (64.3%)
Network Type	Cat	Mobile Data (57.7%), WiFi
Class Duration	Cat	0, 1–3 (69.7%), 3–6 hrs
Self-LMS	Bin	No (82.6%), Yes
Device	Cat	Mobile (84.1%), Computer, Tab
Adaptability Level	Target	Low (39.8%), Moderate (51.9%), High (8.3%)

B. Evaluation metrics

The following metrics were used: Precision, Recall, F1-score, and accuracy are the most important characteristics used to assess the performance of the model.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

For multi-class classification, we employ both macro-averaging which computes unweighted means across classes and weighted-averaging (weighted by the number of true instances per class).

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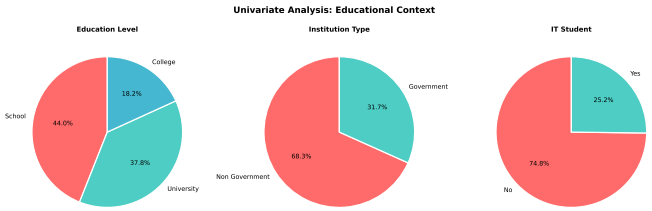


Fig. 2. Univariate Analysis: Educational Context

C. EDA Results

1) Educational context : The sample comprises young students from private schools with limited digital skills and poor resources, struggling with online learning Fig. 2.

2) Technological Infrastructure Analysis: Students primarily used mobile phones with cellular data, facing limited device capability and unstable connections. Most of the learning sessions lasted 1-3 hours, with infrastructure constraints posing obstacles to engagement with online platforms, as illustrated in Fig. 3.

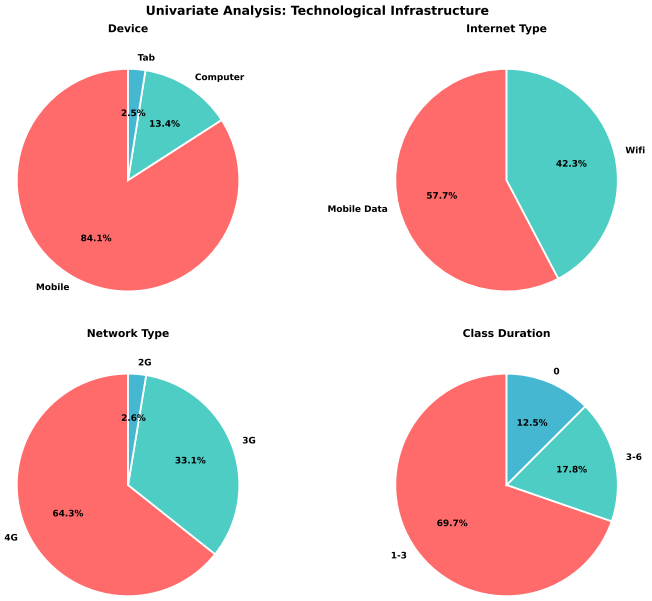


Fig. 3. Univariate Analysis: Technological Infrastructure

D. Feature Engineering Results

- **Tech-Access Index:** Combination of Internet Type and Device Type.
- **Socio-Economic Status Index:** Integration of Financial Condition and Location.
- **Engagement Profile:** Merging Education Level and Self-LMS Usage.

Feature engineering enables the model to learn context-aware learning, revealing the true meaning behind feature interactions, for instance:

- **School + No LMS** A student who needs guidance, which is normal for their age.
- **Mobile + 2G** Students have poor adaptability as they face digital exclusion and connectivity issues.
- **Rural + Poor** Students living in rural areas with financial difficulties. They face complete educational isolation due to their remote location and lack of money.
- **Urban + Poor** Reflects financial constraints only, with educational infrastructure still accessible.

Unlike previous research that used the original raw features without creating interaction between features [3], [4]

E. KmeansSMOTE Result

Table II presents the distribution before and after applying KMeansSMOTE.

TABLE II
CLASS DISTRIBUTION BEFORE AND AFTER KMEANSSMOTE

Metric	Before	After
Samples	1,205	1,505
Class: Low	480 (39.8%)	504
Class: Moderate	625 (51.9%)	500
Class: High	100 (8.3%)	501
Features	16	16

F. Confusion Matrix and Per-Class Performance

We applied KMeansSMOTE on the training dataset after the train-test split to avoid data leakage. Our proposed model achieves 92.9% accuracy, with enhanced minority class detection: High(95% recall), Low(95% recall), and Moderate(91% recall).

Table III compares our results with prior studies. While Suzan et al. [4] achieved 89.63% and Nnadi et al. [5] reached 91%, both omitted class imbalance handling, resulting in relatively low recall recall for the minority (High adaptability) class (approximately 65%). Our framework achieves higher recall for the High class (approximately 95%), suggesting that the integration of class balancing, feature engineering, and hyperparameter optimization improves the model's ability to detect minority class instances.

overall accuracy to 92.9% with minority class detection (95% recall), demonstrating that class balancing enables the model to recognize compound features rather than isolated demographic factors.

The confusion matrix Fig. 4 shows strong diagonal dominance, indicating effective class balance and contextual feature recognition.

G. Benchmark Comparison

Table IV compares models under the same conditions (5-fold CV, same train/test split, 200 Bayesian trials).

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TABLE III
COMPARATIVE PERFORMANCE OF RF MODEL AFTER KMEANSSMOTE

Study	Class	Acc.	Prec.	Rec.	F1	Overall
Suzan et al. (2021) [4]	Low	89.63%	0.89	0.94	0.92	89.63%
	Moderate	—	0.90	0.90	0.90	
	High	—	0.88	0.65	0.75	
Nnadi et al. (2024) [5]	Low	—	0.93	0.92	0.92	91.0%
	Moderate	—	0.90	0.85	0.87	
	High	—	0.88	0.91	0.89	
Efendi et al. (2025) [10]	Low	93%	—	94%	94%	91.3%
	Moderate	90%	—	94%	92%	
	High	88%	—	65%	75%	
Our Study	**Low**	**94.8%**	**0.95**	**0.95**	**0.95**	**92.9%**
	Moderate	**91.2%**	**0.91**	**0.94**	**0.92**	
	High	**95.0%**	**0.83**	**0.95**	**0.88**	

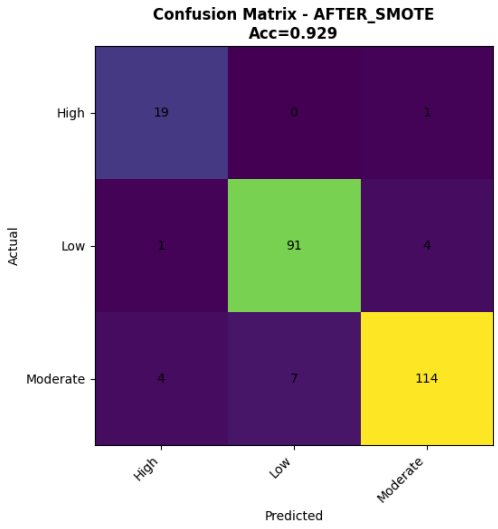


Fig. 4. Confusion matrix after KMeansSMOTE (92.9% accuracy)

TABLE IV
BENCHMARK COMPARISON (5-FOLD CV)

Model	Accuracy	High Recall	Macro F1
RF (No SMOTE)	92.12%	90%	89.9%
XGBoost (No SMOTE)	92.95%	70%	88.8%
XGBoost + SMOTE	92.12%	70%	88.2%
RF + SMOTE (Ours)	**92.9%**	**95%**	**91.2%**

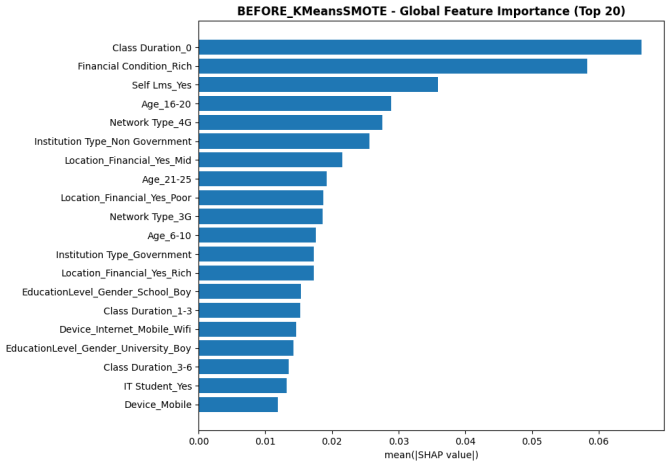


Fig. 5. SHAP Feature importance before KMeansSMOTE

H. SHAP Analysis

1) SHAP before class balancing: According to Fig. 5, Students with zero class duration **Class Duration_0** show the strongest negative impact on student adaptability. The model focused on isolated factors: class duration, financial condition, LMS usage, and network type. The model checks whether students participate in learning and what technology they have.

2) SHAP after class balancing: Figure 6 presents the importance of SHAP features after applying KMeansSMOTE. *Financial Condition_Mid* became the top predictor (mean SHAP value = 0.09), followed by the interaction feature *EducationLevel_Gender_University_Boy* (0.07). *Class Duration_0* remained important (0.05), while engineered feature *Device_Internet_Mobile_Wifi* rose to fourth place (0.02). *Network Type_3G* and *Location_Financial_Yes_Mid* completes the top six features, each with a mean SHAP value of 0.01.

3) Comparative analysis: SHAP (Shapley Additive Explanations) is a game-theoretic framework that assigns a contribution score to each attribute based on Shapley values from game theory [16]. It provides both global and local interpretability with theoretical consistency and fairness [16], [17].

In Educational predictive modeling, explainable AI (XAI) is essential to ensure transparency, fairness, and stakeholder trust [8].

Before oversampling, the model focused on single factors like financial condition and class duration, with some demographic information such as age and institution type. While interaction features that combine multiple factors did not appear among the top predictors.

After KMeansSMOTE, financial condition became the top predictor. Class duration remained highly important. Most notably, the interaction between device type and internet access jumped from rank 17 to 4, showing that balancing helped the model recognize the combined effect of technology factors. Other interaction features, such as the level of education combined with gender, also gained prominence. In the top 10 features, the demographic features decreased from 4 before oversampling to 2 after, so the KMeansSMOTE reduced the heavy reliance on fixed student characteristics, which means KMeansSMOTE reduced demographic bias. The number of features with negligible importance (mean $|\text{SHAP}| < 0.01$) increased from 12 before to 14 after, which is called noise reduction.

KMeansSMOTE improves not only predictive performance but also the model's ability to capture interaction-based relationships. Instead of relying on isolated features, the model focuses on real-world combinations, like

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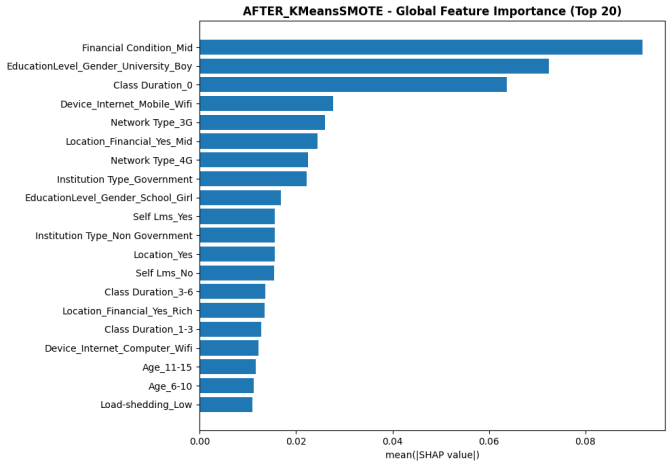


Fig. 6. SHAP feature importance after KMeansSMOTE

financial status with location or device access with internet connectivity. As a result, The engineered interaction features-Tech-access index, Socio-economic Status Index, and Engagement Profile- emerge among the most influential predictors after balancing.

SHAP analysis provides actionable insights for decision-makers in online learning. Students with low class duration and poor financial conditions are consistently associated with low adaptability, indicating that improving engagement time and providing financial or technological support can improve learning outcomes.

Interaction features such as Device_Internet_Mobile_Wifi indicate that improving both device access and connectivity simultaneously is more effective than addressing them independently. These findings support targeted policy interventions in resource-constrained environments.

V. Conclusion and Future Work

This study demonstrates high accuracy in predicting student adaptability levels through the integration of exploratory data analysis, feature engineering, and hyperparameter optimization using a Bayesian algorithm. Unlike previous studies that applied algorithms to raw data, interaction features were engineered, and SHAP interpretations were visualized before and after class balancing using KMeansSMOTE. Feature importance shifts after balancing with some interaction features, gaining prominence while core predictors like financial condition and class duration remain highly important which provide crucial insights for educational policy interventions. Our proposed pipeline achieves 92.9% accuracy using RF classifier with KMeansSMOTE and Bayesian optimization, outperforming manual tuning approaches.The framework prioritizes interpretability with SHAP analysis identifying actionable features such as financial condition, class duration, and the interaction between device type and internet access. A limitation of this paper is the use of a single-country dataset. Future work will extend the model to multi-country datasets to enhance its generalizability in educational environments. Hybrid tree-based and deep learning ensembles can enable real-time adaptive monitoring and personalized interventions.

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